
Beyond Aggregate Accuracy: Fine-Grained Error Analysis and Prompt Sensitivity in LLM Mathematical Reasoning

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Abstract

1 Large language models (LLMs) exhibit impressive yet unreliable mathematical rea-
2 soning. Standard evaluations that focus solely on aggregate accuracy mask critical
3 error patterns, dataset artifacts, and evaluation noise. In this work, we present a
4 fine-grained empirical analysis of LLM errors on GSM8K and the Algebra subset of
5 MATH using two Qwen model variants (Qwen-Plus and Qwen-Turbo) under multi-
6 ple prompting strategies. We first conduct a systematic error taxonomy on GSM8K,
7 identifying interpretation ambiguity and over-reasoning as dominant failure modes
8 and exposing seven dataset-level annotation errors that are misattributed to the
9 model. We then design an iterative re-evaluation protocol and find that prompt-
10 based error recovery is highly unstable: only 4.1% of initially wrong answers are
11 consistently fixed, while over 80% never recover, revealing substantial stochastic
12 variability that undermines single-run evaluations. Furthermore, cross-benchmark
13 and cross-model experiments demonstrate that prompting efficacy is not universal
14 but jointly determined by model capability and task characteristics; strong models
15 achieve near-perfect accuracy with prompt saturation, whereas weaker models can
16 suffer from cognitive overload with few-shot exemplars on multi-step algebraic
17 tasks. Our findings caution against overinterpreting single-run accuracy differences
18 and advocate for stability-aware, dataset-audited evaluation protocols in LLM
19 mathematical reasoning research.

20 1 Introduction

21 Large Language Models (LLMs) have demonstrated remarkable capabilities across a wide spectrum
22 of tasks, yet mathematical reasoning remains a notable challenge. Compared with natural language
23 understanding tasks, mathematical problem-solving demands precise multi-step logical deduction,
24 accurate numerical computation, and faithful adherence to problem constraints Cobbe et al. [2021],
25 Hendrycks et al. [2021].

26 Despite significant progress brought by techniques such as Chain-of-Thought (CoT) prompting Wei
27 et al. [2022] and its zero-shot variant Kojima et al. [2022], many existing evaluations primarily focus
28 on aggregate accuracy metrics, leaving the underlying error mechanisms insufficiently examined.

29 A critical yet underexplored question is: *to what extent do model failures reflect genuine reasoning*
30 *deficiencies versus artifacts of the evaluation process itself?* Recent studies have highlighted that
31 benchmarks such as GSM8K may contain noisy annotations and ambiguous formulations that
32 confound assessment Kim et al. [2024], Mocanu et al. [2024], including incorrect ground-truth
33 answers and underspecified constraints. Moreover, the sensitivity of LLM outputs to prompt variations

34 and the stochastic nature of generation introduce additional complexity rarely accounted for in single-
35 run evaluations, raising concerns about the reliability of current benchmarking practices.

36 In this work, we present a comprehensive empirical study that goes beyond aggregate accuracy to
37 investigate the *quality, stability, and fixability* of LLM mathematical reasoning. We evaluate the Qwen
38 model family on two widely-used benchmarks—GSM8K and MATH Hendrycks et al. [2021]—under
39 three prompting strategies: Direct prompting, Zero-shot CoT, and Few-shot CoT. Our experimental
40 design incorporates an iterative re-evaluation protocol, where incorrectly answered questions are
41 repeatedly fed back to the model with augmented prompts, enabling a fine-grained analysis of error
42 recovery patterns.

43 Our experiments yield several key findings. First, a notable portion of GSM8K errors stem from
44 dataset-level issues—semantic ambiguity, underspecified formats, and contradictions between prob-
45 lem statements and reference solutions—that are erroneously penalized as model failures. Second, the
46 stability of error recovery varies markedly across error categories, with some errors being consistently
47 fixed while others exhibit high stochastic variability across repeated trials. Third, prompting efficacy is
48 jointly modulated by model capability and task characteristics: stronger models (Qwen-Plus) achieve
49 near-saturation on MATH-Algebra regardless of prompt choice, while weaker models (Qwen-Turbo)
50 benefit from CoT prompting yet remain vulnerable to formatting errors.

51 Our main contributions are as follows:

- 52 • We conduct a systematic error attribution analysis on GSM8K, revealing that a non-negligible
53 proportion of model failures stem from dataset-level issues—including semantic ambiguity,
54 potentially incorrect annotations, and underspecified format requirements—rather than pure
55 reasoning deficiencies.
- 56 • We propose an iterative re-evaluation protocol to assess the fixability and stability of different
57 error categories under targeted prompt augmentation. Our results reveal significant stochastic
58 variability in model recovery, suggesting that single-run evaluations may yield unreliable
59 estimates of model performance.
- 60 • Through comparative experiments across two benchmarks and two model scales, we demon-
61 strate that prompting strategy efficacy is jointly determined by model capability and task
62 characteristics.

63 2 Related work

64 2.1 Mathematical reasoning in LLMs

65 GSM8K Cobbe et al. [2021] and MATH Hendrycks et al. [2021] are standard benchmarks for LLM
66 mathematical reasoning, comprising grade-school word problems and competition-level problems
67 across algebra, geometry, and number theory, respectively. More recently, in-context learning
68 strategies have shown that LLMs can achieve competitive performance without parameter updates
69 through carefully designed prompts Brown et al. [2020].

70 2.2 Prompting strategies for reasoning

71 Chain-of-Thought (CoT) prompting Wei et al. [2022] improves LLM mathematical performance by
72 providing exemplars with explicit step-by-step reasoning. Building on this, Kojima et al. Kojima
73 et al. [2022] showed that a simple zero-shot instruction—*Let’s think step by step*—can elicit similar
74 reasoning behavior without demonstrations, a technique known as Zero-shot CoT. Wang et al.

75 While these methods have been extensively evaluated in terms of overall accuracy, fine-grained
76 analyses of *which* error types are amenable to prompting interventions remain scarce. Our work
77 addresses this gap by systematically categorizing errors and studying their fixability under different
78 prompting conditions.

79 2.3 Evaluation challenges and benchmark limitations

80 Benchmark annotation errors and format ambiguity have been noted as confounding factors Kim
81 et al. [2024], Mocanu et al. [2024], Jo et al. [2025]. Answer extraction in math benchmarks—parsing

free-form outputs into final answers—introduces evaluation noise, particularly in MATH where diverse symbolic representations (e.g., $\frac{1}{2}$ vs. 0.5) cause correct reasoning to be marked incorrect Jo et al. [2025], Hendrycks et al. [2021]. Furthermore, the inherent stochasticity of LLM generation means that single-run evaluations may not reliably capture model capabilities Potamitis et al. [2025], Atil et al. [2025]. Our work contributes by explicitly modeling format sensitivity and output stability as important evaluation dimensions, rather than treating them as negligible noise.

3 Methodology

3.1 Task formulation

We formulate mathematical reasoning as a conditional sequence generation task. Given a math problem text x , the Large Language Model (LLM) is tasked with generating a response $\hat{y}$ that ideally contains a reasoning trace followed by a final answer. Let y^* denote the ground-truth answer for x .

The primary evaluation metric is the answer-level accuracy, defined as:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\text{Match}(\hat{y}_i, y_i^*)) \quad (1)$$

where N is the total number of evaluated questions, $\mathbb{I}(\cdot)$ is the indicator function, and $\text{Match}(\cdot, \cdot)$ is our custom equivalence judgment function detailed in Section 3.2.

3.2 Answer extraction and evaluation

Accurate answer extraction and matching are essential for math reasoning evaluation, yet they are often overlooked or oversimplified in existing benchmarks. To ensure full reproducibility and minimize evaluation noise, we design separate, strictly rule-based pipelines for GSM8K and MATH without relying on external symbolic solvers or auxiliary LLMs.

GSM8K. The official format is ##### number. We prompt models to follow this convention. We extract the last numeric token after the final ##### marker (or from the full response if absent), strip formatting symbols (\$, commas), and compare with the ground truth via exact numerical equality. Responses without an identifiable answer are treated as incorrect.

MATH. Answers are LaTeX expressions in `\boxed{\}`. Our pipeline comprises three stages: (1) *localization*—extract the last `\boxed{\}` via brace-counting, falling back to explicit markers, last-line, or numeric extraction; (2) *normalization*—remove LaTeX formatting commands, unify fraction notations, and strip variable prefixes; (3) *equivalence judgment*—a cascade of exact match, normalized match, float comparison (10^{-6} tolerance), and fraction-aware comparison. Full regex specifications and the complete cascade are provided in Appendix B. *(There are very few format blind spots in the automatic extraction, which will be supplemented by manual review in Section 4.)*

3.3 Prompt strategies

Inspired by prior work on Chain-of-Thought (CoT) reasoning Wei et al. [2022], Kojima et al. [2022], we evaluate three prompting strategies on the first 300 questions of the GSM8K and MATH test sets to isolate the impact of reasoning elicitation. All prompts are appended as instructions to the end of the original problem text. (Full prompt templates are provided in Appendix A).

- **Zero-shot CoT:** Guides the model to reason step-by-step without providing examples. For GSM8K, it explicitly requests the ##### format, forbids thousand separators, and includes a reminder for unit conversions. For MATH, it mandates the use of `\boxed{\}` and LaTeX formatting.
- **Direct Prompting (No CoT):** Removes the step-by-step reasoning instruction, directly requesting the final answer in the specified format. This serves as our baseline to measure the pure gain from CoT.
- **Few-shot CoT:** Replaces the verbal reasoning instruction with 5 in-context examples drawn from outside the 300-question evaluation set to prevent data leakage.

- 126 – *For GSM8K*: To ensure diverse coverage, we manually categorize the problem types
127 and select one error-free exemplar with its official solution for each category.
- 128 – *For MATH*: Since the first 300 questions are exclusively algebra problems, we randomly
129 sample 5 algebra problems with official solutions as exemplars.

130 Each exemplar consists of a problem statement and a complete standard reasoning process,
131 preceded by an instruction to learn from the examples.

132 3.4 Iterative re-evaluation protocol

133 To systematically investigate the fixability of specific error types and the output stability of mathe-
134 matical reasoning, we design an iterative re-evaluation protocol, applied specifically to the GSM8K
135 dataset.

136 The protocol initializes with the Zero-shot CoT prompt (excluding the unit conversion reminder) and
137 conducts multiple evaluation rounds on the same problem set. Let $\mathcal{E}_t$ denote the set of incorrectly
138 answered questions at round t .

139 Procedure:

- 140 1. *Initial Evaluation* ($t = 0$): Run the model on all problems using the base Zero-shot CoT
141 prompt. Determine correctness via Section 3.2 to obtain the initial error set $\mathcal{E}_0$.
- 142 2. *Prompt Refinement*: For each problem in $\mathcal{E}_t$, keep the problem text unchanged but append
143 a targeted reminder to the prompt: “*and pay attention to unit conversions.*” The model
144 weights remain strictly identical.
- 145 3. *Re-evaluation*: Re-run the model exclusively on the problems in $\mathcal{E}_t$ using the refined prompt,
146 extract new answers, and re-judge correctness to obtain $\mathcal{E}_{t+1}$.
- 147 4. *Iteration*: Repeat steps 2–3 for T additional rounds. In this work, we set $T = 3$, resulting in
148 a total of 4 runs (1 initial + 3 re-evaluations).

149 **Reporting convention:** The objective of this protocol is to observe performance fluctuations as the
150 prompt is incrementally strengthened. Therefore, we report the accuracy and error composition of
151 each round independently. This experimental design allows us to cleanly disentangle two distinct
152 phenomena: the intrinsic stochastic variability of the LLM (observable when the same prompt yields
153 different results across runs) and the systematic performance gain derived from targeted prompt
154 reinforcement.

155 4 Experiments and results

156 4.1 Implementation details

157 All experiments are conducted via the official Qwen API using the `Generation.call` interface. We
158 evaluate two model scales: Qwen-Plus and Qwen-Turbo. Unless otherwise specified, we adopt the
159 API’s default decoding configuration to assess the models’ out-of-the-box capabilities: temperature
160 $= 0.7$, $\text{top-}p = 0.8$.

161 4.2 RQ1: error categorization and fixability on GSM8K

162 4.2.1 Baseline performance

163 We first establish the baseline performance of Qwen-Plus on the full GSM8K test set (1,319 questions)
164 using the Zero-shot CoT prompt (excluding the unit-conversion reminder). The model achieves
165 an accuracy of 96.29% (49 errors), providing a solid foundation for subsequent fine-grained error
166 analysis.

167 4.2.2 Error distribution and dataset-level issues

168 We manually categorize the 49 incorrect predictions into six error types. As Figure 3 in Appendix C
169 illustrates, *Interpretation Ambiguity* and *Over-reasoning* constitute the largest proportions, indicating
170 that the model’s errors primarily stem from misinterpretation of problem statements rather than

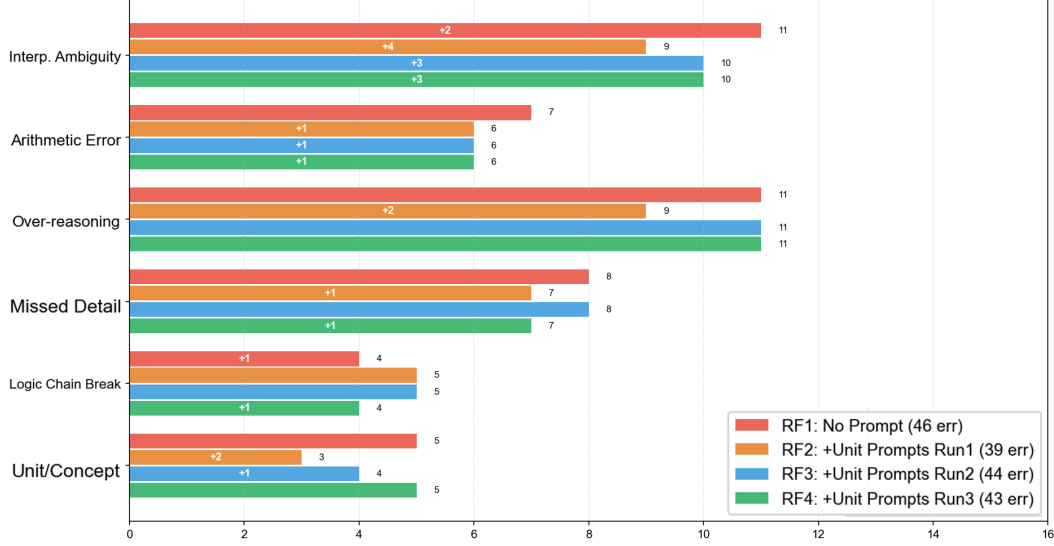


Figure 1: Per-category error counts across four evaluation rounds on the 49 incorrectly answered GSM8K problems. RF1 re-runs the original prompt; RF2–RF4 append a unit-conversion reminder. The non-monotonic trajectory (46→39→44→43) indicates that prompt augmentation yields inconsistent, run-dependent gains.

computational deficiencies. To some extent, this also reflects inherent semantic ambiguity within certain questions in the dataset itself. More critically, during manual inspection we identify **7 problems** (IDs: 256, 455, 824, 953, 1039, 1043, 1310) for which the ground-truth solution is fundamentally inconsistent with the problem statement. We independently recomputed all 7 problems and confirmed that the model’s outputs are mathematically correct. These cases are reclassified into a distinct *Dataset Annotation Error* category (see Figure 7 in Appendix D for all the cases). This finding corroborates recent observations on GSM8K annotation quality and underscores that a non-trivial fraction of reported “model errors” are in fact evaluation artifacts.

4.2.3 Iterative re-evaluation: non-monotonic gains

To investigate whether targeted prompt augmentation can repair specific error types, and whether such repairs are stable, we apply the Iterative Re-evaluation Protocol described in Section 3.4. Specifically:

- RF1 (Control re-run):** The 49 incorrectly answered questions are re-fed to the model with the *identical* Zero-shot CoT prompt (no modification). This establishes the stochastic baseline for error recovery.
- RF2–RF4 (Prompt augmentation):** A unit-conversion reminder (“*and pay attention to unit conversions*”) is appended to the prompt. The model is re-evaluated on the same 49 questions for three consecutive runs.

As shown in Figure 1, the control re-run (RF1) reduces errors from 49 to 46, confirming the presence of stochastic variability even without prompt modification. Appending the unit-conversion reminder (RF2–RF4) further reduces errors to 39–44, but the improvement is **non-monotonic and run-dependent**: RF2 achieves the best result (39 errors), yet RF3 regresses to 44 before partially recovering in RF4 (43). At the category level, the *Unit/Concept* errors show the largest relative reduction (from 5 to 3–4), suggesting that explicit reminders can marginally mitigate format-related confusion. However, categories such as *Interpretation Ambiguity* and *Over-reasoning* remain largely invariant, indicating that these errors are structurally entrenched and insensitive to surface-level prompt augmentation.

197 4.2.4 Recovery stability: the rarity of consistent fixes

198 We further analyze the *stability* of error recovery across the three prompt-augmented runs (RF2–RF4).
199 See Figure 4 in Appendix C for details.

200 The high degree of stochastic variability carries an important methodological implication: **observing**
201 **a “fix” in a single re-run may be a spurious signal rather than evidence of genuine capability**
202 **improvement**. Our results underscore the necessity of multi-run stability analysis in error attribution
203 studies and validate the design choice in Section 3.4 to report each round independently rather than
204 aggregating via union or intersection.

205 4.3 RQ2: cross-model, cross-task prompting analysis

206 4.3.1 GSM8K (Q1–300): Marginal Prompt Sensitivity

207 We compare three prompting strategies (Zero-shot CoT, Direct Prompting, Few-shot CoT) on a
208 controlled subset of the first 300 questions of the GSM8K test set, using Qwen-Plus.

209 As shown in Figure 5 in Appendix C, Zero-shot CoT and Few-shot CoT achieve *identical* accuracy
210 (95.7%, 13 errors each), while Direct Prompting (No-CoT) trails marginally at 95.0% (15 errors).
211 The negligible inter-strategy gap indicates that, for a capable model on a relatively simple benchmark,
212 prompt engineering yields diminishing returns—a phenomenon we term *prompt saturation*. Crucially,
213 **6 questions are incorrectly answered by all three strategies** (see Figure 8 in Appendix E.1 for
214 the detailed comparison). Manual inspection reveals that this shared failure set includes Q256, a
215 confirmed dataset-level annotation error (Section 4.2.2), along with several problems exhibiting
216 severe semantic ambiguity. The fact that even the strongest prompting configuration cannot resolve
217 these cases suggests that their failures are *fundamentally irrecoverable via prompt engineering*. An
218 additional observation merits attention: although Few-shot CoT and Zero-shot CoT achieve identical
219 overall accuracy, their *error distributions differ*—Few-shot fixes 6 questions that Zero-shot misses,
220 while regressing on 6 others. (see Figure 8 in Appendix E.1) This **zero-sum redistribution** indicates
221 that in-context exemplars primarily shift the model’s reasoning style rather than uniformly improving
222 reasoning quality.

223 4.3.2 MATH-Algebra (Q1–300): capability-dependent ceiling effects

224 We apply the same three prompting strategies to the first 300 questions of the MATH test set (all
225 belonging to the Algebra category), evaluating both Qwen-Plus and Qwen-Turbo to examine how
226 model capability modulates prompt effectiveness.

227 **(a) Qwen-Plus: Near-Perfect Saturation.** As illustrated in Figure 6 in Appendix C, Qwen-Plus
228 achieves a raw accuracy of 98% on MATH-Algebra, with virtually no difference across the three
229 prompting strategies. The flat accuracy curve (bottom-right panel) indicates that the model’s intrinsic
230 algebraic reasoning capacity *saturates* this subset—a clear ceiling effect. Moreover, all 6 errors
231 across all runs are attributable to **format mismatches** rather than reasoning failures (e.g. predicted
232 $7(x-3)(x+3)$ vs. gold $7(x+3)(x-3)$). While Few-shot CoT does not improve accuracy, it
233 marginally increases the exact-match count (from 258 under Zero-shot to 260 under Few-shot),
234 suggesting that in-context exemplars help the model internalize the benchmark’s preferred formatting
235 conventions.

236 **(b) Qwen-Turbo: Non-Monotonic Prompt Effects.** Qwen-Turbo exhibits a **counterintuitive**
237 performance ranking: Zero-shot CoT (96.0%) > Direct Prompting (95.0%) > Few-shot CoT (94.7%)
238 in terms of raw accuracy. The top panel of Figure 2 highlights a critical distinction between raw
239 and real accuracy. After our manual inspection, we eliminate the format mismatches and got the
240 actual accuracy of the model: Zero-shot CoT (98.7%) > Direct Prompting (97.7%) > Few-shot CoT
241 (97.0%). The 2–4% gap between raw and real accuracy quantifies the prevalence of format-induced
242 false negatives and validates our evaluation design. Surprisingly, Few-shot CoT *underperforms* on
243 Qwen-Turbo. We attribute this to **cognitive overload**: the 5 in-context algebra exemplars consume
244 the model’s limited context capacity, leaving fewer computational resources for the target problem’s
245 symbolic manipulation. The bottom-left panel of Figure 2 supports this hypothesis: Few-shot incurs
246 9 reasoning errors compared to only 4 under Zero-shot CoT, despite having comparable format error
247 counts. Figure 10 in Appendix E.3 lists the reasoning errors that persist across prompting strategies.

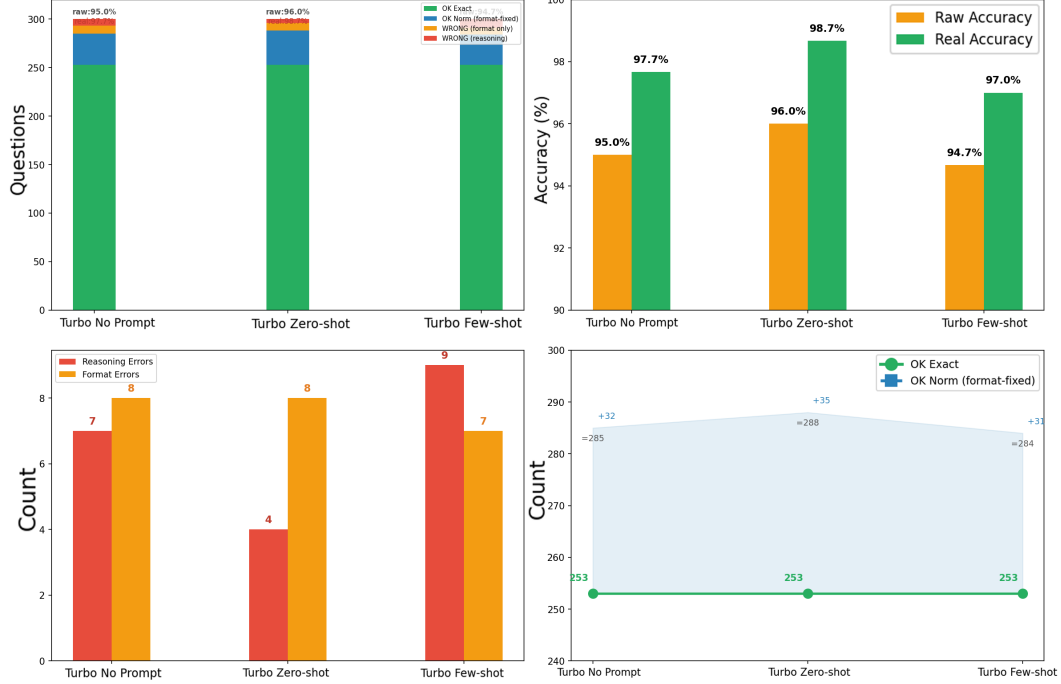


Figure 2: MATH-Algebra (Q1–300) evaluation results for Qwen-Turbo. **Top-left:** Answer matching distribution by strategy. **Top-right:** Raw vs. real accuracy—normalization recovers 2–4% of format-induced false negatives, revealing the true capability hierarchy (Zero-shot > No-CoT > Few-shot). **Bottom-left:** Format vs. reasoning error decomposition—Few-shot incurs 9 reasoning errors vs. only 4 under Zero-shot CoT, despite comparable format error counts. **Bottom-right:** Exact vs. normalized match counts per strategy. The non-monotonic ranking across strategies reveals cognitive overload effects in weaker models.

At first glance, our results appear contradictory: on GSM8K, Few-shot CoT matches Zero-shot CoT; on MATH-Algebra, it underperforms. These observations can be **unified by a single principle**: *prompting efficacy is jointly determined by model capability and task characteristics*. The non-monotonicity challenges the common assumption that “more prompting always helps” and suggests that **prompt engineering must be calibrated to the specific model–task pair** rather than applied uniformly.

5 Conclusion

In this work, we presented a fine-grained empirical study of error behavior and prompt sensitivity in LLMs on mathematical reasoning benchmarks. Our three main contributions are: (i) a taxonomy revealing interpretation ambiguity and over-reasoning as dominant failure modes, plus 7 dataset-level annotation errors in GSM8K misattributed to models; (ii) an iterative re-evaluation protocol showing that prompt-based fixes are highly unstable—only 4.1% of errors are consistently recovered, while 81.6% never recover; (iii) a non-monotonic interaction between model capability and prompt complexity, where Few-shot CoT underperforms for weaker models on structured tasks due to cognitive overload. These findings suggest that prompting efficacy must be calibrated to the model–task pair rather than applied uniformly, and that multi-run stability analysis should be a methodological prerequisite in error attribution studies. We acknowledge limitations in scope (two model scales, two datasets), reliance on a proprietary API, and the correlational nature of our cognitive overload hypothesis. Future work includes extending this framework to open-weight models and developing automated benchmark auditing tools.

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A Prompt Templates

In this section, we provide the exact prompt templates used in our experiments. For the GSM8K dataset, the original prompts used in our experiments were in Chinese to align with the model’s primary instruction-tuning language; however, we provide their English translations here for clarity and reproducibility. For the MATH dataset, all prompts were in English.

A.1 GSM8K Prompts

A.1.1 Zero-shot CoT Prompt

The following instruction is appended to the end of the original problem text:

```
\n\nPlease reason step by step, and write the final answer on a
separate line at the end, formatted as '#### answer_number'. Note
that the answer number should be a pure number without separators
such as 8,000, and pay attention to unit conversions.
```

Note: In the initial round of the Iterative Re-evaluation Protocol (Section 3.4), the clause “and pay attention to unit conversions” is omitted.

311 A.1.2 Direct Prompting (No CoT)

312 The reasoning instruction is removed, and the following is appended instead:

313 \n\nPlease write the final answer on a separate line at the end,
314 formatted as '#### answer_number'. Note that the answer number
315 should be a pure number without separators such as 8,000, and pay
316 attention to unit conversions.

317 A.1.3 Few-shot CoT Prompt ($k = 5$)

318 The verbal reasoning instruction is replaced with the following example-learning instruction and 5
319 exemplars:

320 Please read the following example questions and learn the reasoning
321 process of the standard answers.

322 Exemplar 1:

323 **Question:** Jen decides to travel to 3 different countries. He has to pay \$400 for the supplies
324 he needs, in total. The tickets for travel cost, in total, 50% more than the supplies. How
325 much does travel cost?
326 **Solution:** He pays $400 \times .5 = \$200$ more for tickets than supplies. That means the tickets cost
327 $400 + 200 = \$600$. So it cost $600 + 400 = \$1000$ in total. ##### 1000

328 Exemplar 2:

329 **Question:** Sasha and Julie are best friends playing on opposing basketball teams. The teams
330 have two practice games scheduled. In the first game, Sasha had the home court advantage
331 and scored 14 points. Julie scored 4 fewer points than Sasha in the same game. Sasha always
332 struggles during away games and their second match was at Julie's home court. Sasha scored
333 6 fewer points in the second game than Julie's score in the first game. How many total points
334 did Sasha score during both games?
335 **Solution:** In the first game, Sasha scored 14 points, and Julie scored four fewer points: $14 - 4$
336 $= 10$ points. In the second game, Sasha scored 6 points fewer than Julie's score in the first
337 game, meaning she scored $10 - 6 = 4$ points. Sasha scored $10 + 4 = 14$ points in the two games.
338 ##### 14

339 Exemplar 3:

340 **Question:** Teddy finished half of a 500 piece puzzle, and then started and finished another
341 500 piece puzzle within an hour. How many puzzle pieces did Teddy place during that hour?
342 **Solution:** Teddy did $1/2 \times 500$ pieces = 250 pieces. Teddy completed 250 pieces + 500
343 pieces = 750 pieces. ##### 750

344 Exemplar 4:

345 **Question:** Chase and Rider can ride their bikes thrice a day for 5 days; but on two other
346 days, they ride twice the times they do on usual days. How many times do they ride their
347 bikes a week?
348 **Solution:** Each person rides his bike $3 \times 5 = 15$ times for 5 days. Together, they ride $15 + 15$
349 $= 30$ times for five days. Each person rides his bike $3 \times 2 = 6$ times for every day of the other
350 two days. This means each person rides $6 \times 2 = 12$ times on each day of the other two days.
351 The total for the other two days is $12 + 12 = 24$. For the whole week the together ride $24 + 30$
352 $= 54$ times. ##### 54

353 Exemplar 5:

354 **Question:** Imma has 3 cats. She feeds her cats twice a day with 60 grams of cat food. How
355 many days will 720 grams of cat food last?
356 **Solution:** The 3 cats consume 60 grams/cat $\times$ 3 cats = 180 grams of food per meal. Since
357 they eat twice a day, then they consume 180 grams/meal $\times$ 2 meals/day = 360 grams in a day.
358 So, 720 grams of cat food only last for $720 \text{ grams} / 360 \text{ grams/day} = 2$ days. ##### 2

359 A.2 MATH Prompts

360 A.2.1 Zero-shot CoT Prompt

361 The following instruction is appended to the end of the original problem text:

362 \texttt{\n\nPlease reason step by step, put your final answer inside
363 \boxed{}}. Use LaTeX for mathematical expressions.

364 A.2.2 Direct Prompting (No CoT)

365 The reasoning instruction is removed:

366 \texttt{\n\nPlease put your final answer inside \boxed{}}. Use LaTeX for
367 mathematical expressions.

368 A.2.3 Few-shot CoT Prompt ($k = 5$)

369 The verbal instruction is replaced with the following instruction and 5 randomly sampled algebra
370 exemplars:

371 Please study the reasoning process of the solutions in the following
372 example questions.

373 Exemplar 1:

374 **Problem:** Evaluate $\lceil 3(6 - \frac{1}{2}) \rceil$.
375 **Solution:** Firstly, $3(6 - \frac{1}{2}) = 18 - 1 - \frac{1}{2} = 17 - \frac{1}{2}$. Because $0 \leq \frac{1}{2} < 1$, we have
376 $\lceil 17 - \frac{1}{2} \rceil = \boxed{17}$.

377 Exemplar 2:

378 **Problem:** Sam is hired for a 20-day period. On days that he works, he earns \$60. For each
379 day that he does not work, \$30 is subtracted from his earnings. At the end of the 20-day
380 period, he received \$660. How many days did he not work?
381 **Solution:** Call x the number of days Sam works and y the number of days he does not. We
382 can set up the following system of equations: $x + y = 20$ and $60x - 30y = 660$. Solving
383 for x in the first equation yields $x = 20 - y$. Substituting into the second equation gives
384 $60(20 - y) - 30y = 660$. Simplifying yields $-9y = -54$, or $y = 6$. Thus, Sam did not
385 work for $\boxed{6}$ days.

386 Exemplar 3:

387 **Problem:** Find the center of the circle with equation $x^2 - 6x + y^2 + 2y = 9$.
388 **Solution:** Completing the square, we get $(x - 3)^2 + (y + 1)^2 = 19$. Therefore, the center
389 of the circle is $\boxed{(3, -1)}$.

390 Exemplar 4:

391 **Problem:** If $x = 2$ and $y = 5$, then what is the value of $\frac{x^4 + 2y^2}{6}$?
392 **Solution:** We have $\frac{x^4 + 2y^2}{6} = \frac{2^4 + 2(5^2)}{6} = \frac{16 + 2(25)}{6} = \frac{16 + 50}{6} = \frac{66}{6} = \boxed{11}$.

393 Exemplar 5:

394 **Problem:** The sequence of integers in the row of squares and in each of the two columns of
395 squares form three distinct arithmetic sequences. What is the value of N ? (Accompanied by
396 an Asymptote diagram).
397 **Solution:** Since $18 - 14 = 4$, the common difference in the first column is 4, so the
398 number above 14 is 10, and above 10 is 6. This is the fourth number in the row, so the
399 common difference in the row is $(6 - 21)/3 = -5$. The seventh number in the row is
400 $21 - 5 \cdot 6 = -9$. In the second column, the common difference is $[(-17) - (-9)]/4 = -2$,
401 so $N = -9 - (-2) = \boxed{-7}$.

B Answer Extraction Details

B.1 GSM8K Extraction

The official GSM8K answer format is ##### number. We prompt models to follow this convention. **Answer localization** proceeds in two stages: if the generated text contains #####, we extract the substring after its last occurrence and strip \$ signs and commas; otherwise, the same cleaning is applied to the entire response. All numeric tokens matching the regular expression $[-+]?[d*]\.?[d+]$ are extracted, and the *last* valid token is selected as the final prediction, converted to an integer or float. If no number is found, the prediction is marked as missing. **Normalization** solely removes commas and \$ signs. **Equivalence judgment** relies on exact numerical equality ($\hat{y} == y^*$). Since GSM8K ground truths are strictly integers or finite decimals, this comparison is unambiguous.

B.2 MATH Extraction

Answers in the MATH dataset are typically LaTeX expressions enclosed in `\boxed{\}`. Our pipeline employs a four-layer localization strategy and a cascaded equivalence check.

Answer localization follows a strict priority order:

1. *Boxed extraction*: Using brace-counting to handle nested structures, we extract the content of the *last* `\boxed{\dots}` in the text.
2. *Explicit marker matching*: If Step 1 fails, we search for trigger phrases (e.g., “the answer is”, “final answer”) and capture the subsequent substring.
3. *Last-line extraction*: The last non-empty line, provided it is shorter than 200 characters, is taken as the candidate.
4. *Numeric fallback*: As a last resort, all LaTeX commands and punctuation are stripped, and the last numeric token is extracted.

The extracted raw string undergoes **normalization** using purely regex and string operations. Key steps include: stripping \$ wrappers; removing formatting commands (e.g., `\text`, `\mathrm`) and their contents; unifying `\dfrac` to `\frac`; removing `\left`, `\right`, and spacing commands; expanding shorthands (e.g., `\frac{1}{2}` to `\frac{1}{2}`); and stripping variable prefixes (e.g., $x =$) and trailing punctuation. Irrational numbers, square roots, and π are preserved in symbolic form without numerical approximation.

Equivalence judgment utilizes a cascade of five strategies, returning True upon the first successful match:

1. Exact string match of raw answers;
2. Exact string match post-normalization;
3. Float comparison with an absolute tolerance of 10^{-6} (skipped if safe float conversion fails);
4. Fraction-aware comparison: parses both `\frac{a}{b}` and `a/b`, converts to float, and applies the 10^{-6} tolerance;
5. Exact match after removing all whitespace.

Handling Missing and Multiple Answers: If the pipeline yields a null answer, the response is scored as incorrect. When multiple `\boxed{\}` instances appear or the model self-corrects during reasoning, the *last* identified answer is always used, assuming the model’s final conclusion appears at the end. Potential biases from this assumption are discussed in our error analysis (Section 4).

C Experiments’ charts

Figure 3 presents the error distributions in the baseline. See Section 4.2.4.

Figure 4 provides the Intuitive stability analysis of the fixed problems. See Section 4.2.

Figure 5 presents the accuracy of the three strategies on GSM8K. See Section 4.3.1.

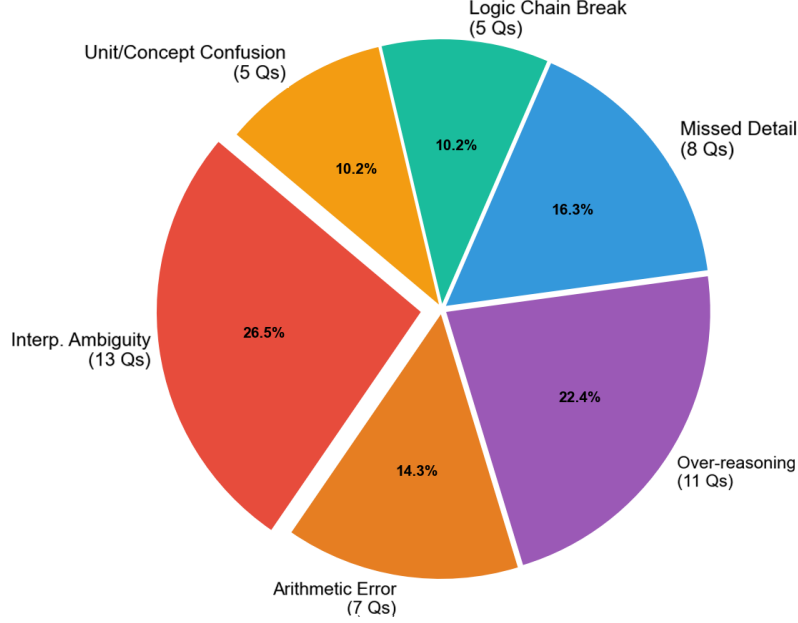


Figure 3: Distribution of error categories across 49 incorrect predictions of Qwen-Plus on the full GSM8K test set under Zero-shot CoT prompting. Interpretation Ambiguity and Over-reasoning dominate, suggesting that dataset-level semantic issues play a non-trivial role.

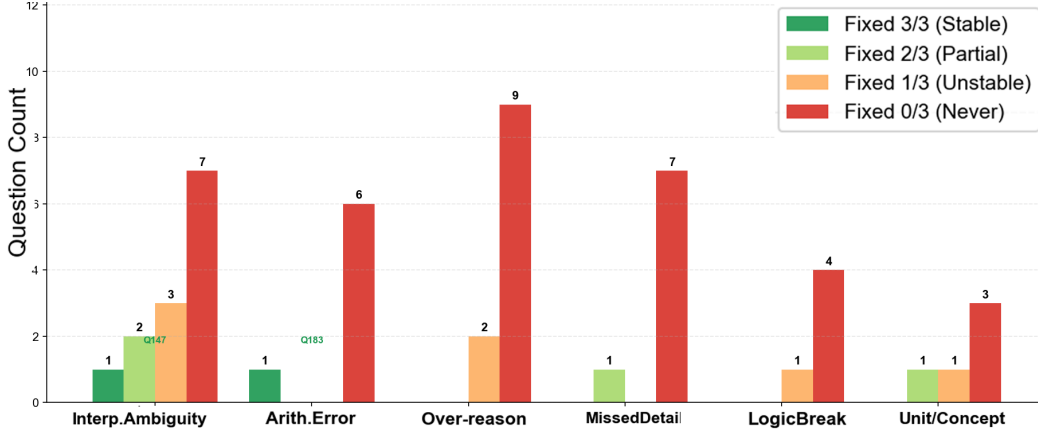


Figure 4: Fix stability distribution across 49 re-evaluated GSM8K problems over three prompt-augmented runs (RF2–RF4). Only 2 questions (4.1%) are consistently fixed in all three rounds, while 40 (81.6%) never recover. The high stochasticity undermines the reliability of single-run re-evaluations.

446 Figure 6 provide an overall analysis of the qwen-plus’s performance using three prompt strategies.
 447 See Section 4.3.2.

448 D Dataset-Level Annotation Errors

449 Figure 7 presents the complete details of the 7 problems in the GSM8K test set for which we
 450 identified inconsistencies between the problem statement and the reference solution. For each case,
 451 we independently recomputed the answer and confirmed that the model’s output is mathematically
 452 correct. These cases are discussed in Section 4.2.2.

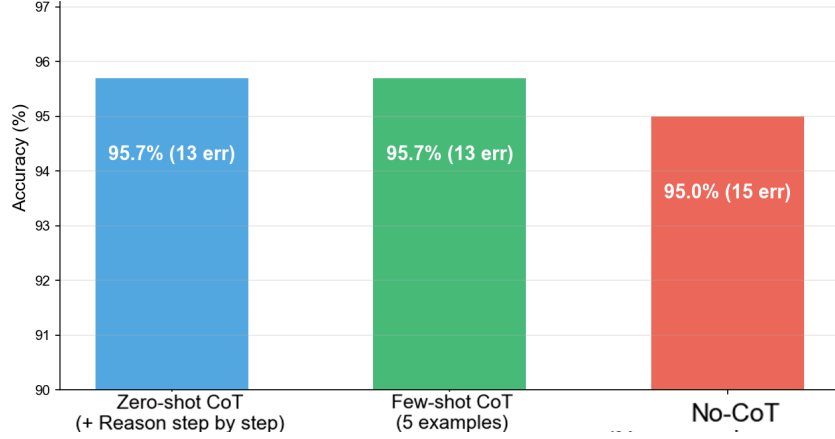


Figure 5: Accuracy of three prompting strategies on GSM8K Q1–300 (Qwen-Plus). Zero-shot CoT and Few-shot CoT achieve identical accuracy (95.7%), while Direct Prompting trails marginally (95.0%), indicating prompt saturation for strong models on simple tasks.

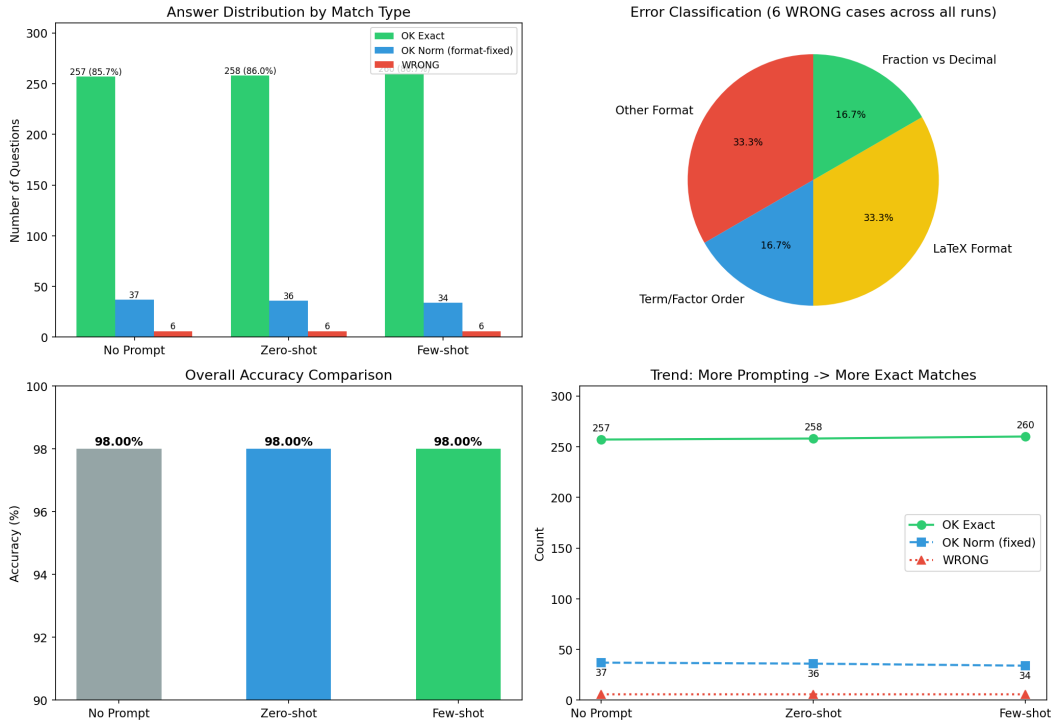


Figure 6: MATH-Algebra (Q1–300) results for Qwen-Plus. Top-left: answer matching distribution. Top-right: error category breakdown (all format-related). Bottom-left: accuracy by strategy (all 98%). Bottom-right: exact-match trend (257→258→260).

453 E Detailed Error Analysis

454 E.1 GSM8K Q1–300: Three-Strategy Error Comparison

455 Figure 8 provides the per-question correctness comparison across all three prompting strategies on
 456 GSM8K Q1–300 (Qwen-Plus). This figure supports the analysis in Section 4.3.1, including the
 457 identification of 6 shared failures and the zero-sum redistribution between Few-shot and Zero-shot
 458 CoT.

Dataset-level annotation errors

ID	Details	Original category
256	The question states clearly that the new cows are to be distributed across 20 stalls, yet the standard answer assumes 10 stalls; $40/10=4$, ignoring the fact that there should be 20 stalls.	Interpretation Ambiguity
455	The question states that "each person eats 4" (literal translation), yet the reference solution computes $4+1=5$;	Arithmetic Error
824	In the standard process, Sasha scored 14 in the first round and 4 in the second round, yet the standard answer was written as $10+4=14$, mistaking Sasha's 14 in the first round for 10. The correct total should be $14+4=18$.	Arithmetic Error
953	The question specifies a time range from 8 a.m. to 11 p.m., but the reference solution incorrectly uses 8 a.m. to 11 a.m.	Missed Detail
1039	The standard answer clearly calculates $300,000 \div 50,000 = 6$ years, but they ended up writing 4, which doesn't make any sense.	Logic Chain Break
1043	Won 11 in the second game, lost 1.5 times that in the third game, so it should be $11 \times 1.5 = 16.5$, but the standard answer says $10 \times 1.5 = 15$, which makes the average loss 3 when it should be 3.5.	Logic Chain Break
1310	The standard answer added 400 twice when summing up at the end: $750+430+400+700$, when it should actually be $430+750+300+700=2180$	Arithmetic Error

Figure 7: Complete details of 7 dataset-level annotation errors identified in the GSM8K test set. For each problem, the table shows the problem ID, an abbreviated problem statement, the model’s answer, the specific inconsistency in the reference solution, and the originally assigned error category. All 7 model outputs were verified to be mathematically correct.

E.2 Four-Round Iterative Re-evaluation: Per-Question Results

Figure 9 reports the per-question outcomes across all four evaluation rounds (RF1–RF4) for the 49 initially incorrect GSM8K questions. This figure supports the stability analysis in Section 4.2.4.

E.3 MATH Turbo: Persistent Reasoning Errors

Figure 10 extends the analysis in Section 4.3.2(b) by listing all persistent reasoning errors of Qwen-Turbo on MATH-Algebra (Q1–300) across the three prompting strategies.

Q#	Category	Error Reason	Zero-shot CoT	Few-shot CoT	No-CoT	UnitFix	Note
Q1	Interp.Amb	two times older than: 3x vs 2x	121=8	121=8	121=8	3	All prompts fail (hard)
Q192	Interp.Amb	start/stop not full (string vs Q)	111=12	111=12	111=12	3	All prompts fail (hard)
Q235	Interp.Amb	ratio/grouping ambiguity	141=21	141=21	141=21	3	All prompts fail (hard)
Q254	Interp.Amb	Ten vs Twenty state confusion	90=112	90=112	90=112	3	All prompts fail (hard)
Q236	Interp.Amb	used spoons: consumed vs still there	OK	OK	OK	2	FS+NC both wrong
Q242	Interp.Amb	collectively read: unique vs sum	OK	OK	OK	2	ZS+NC both wrong
Q234	Interp.Amb	2 times more + 4th round confusion	1041=83	1041=83	OK	2	ZS+FS both wrong
Q232	Interp.Amb	2 times more: 3x vs 2x	75=150	OK	OK	1	Only ZS wrong (FS+NC OK)
Q154	Arith.Err	add cost value vs add	54=102	54=102	102=102	3	All prompts fail (hard)
Q29	Arith.Err	subtraction/comparison error	OK	87=127	87=127	2	FS+NC both wrong
Q16	Arith.Err	age boundary: 34, 20+1 overcounts	48=44	OK	48=44	2	ZS+NC both wrong
Q155	Arith.Err	basic arithmetic error	OK	100=105	OK	3	FS+NC both wrong
Q289	Arith.Err	multi-step arithmetic mistake	4=3	OK	OK	1	Only ZS wrong (FS+NC OK)
Q181	Over-reas	over-complicated ratio problem	750=175	OK	750=175	2	ZS+NC both wrong
Q188	Over-reas	simple vs compound interest	OK	158, 121=158	158, 121=158	2	FS+NC both wrong
Q216	MissedDetail	missed person in growing count	OK	102=140	OK	1	Only FS wrong (ZS+NC OK)
Q299	MissedDetail	total/grouping count error	OK	400=1000	OK	1	Only FS wrong (ZS+NC OK)
Q238	Logic/Inv	salary chain computation broken	96077=192000	96077=192000	96077=192000	3	All prompts fail (hard)
Q14	UnitConf	precision: 401 = 100.30 vs 30	OK	OK	OK	2	ZS+NC both wrong
Q179	UnitConf	magnitude error: 1200 vs 95	OK	OK	1200=95	1	Only NC wrong (ZS+FS OK)

Accuracy: Zero-shot CoT=95.7%(13err) | Few-shot CoT=95.7%(13err) | No-CoT=95.0%(15err)

Shared errors (3/3 all wrong): 7 hard Qs - Q41,120,162,164,235,256, etc.

Few-shot fixed vs ZS: Q86,94,101,232,242,289 (6 Qs) | Few-shot regressed: Q59,126,185,188,210,299 (6 Qs)

No-CoT adds 2 more errors (Q59,175) vs ZS. Removing CoT is harmful.

Conclusion: Few-shot = Zero-shot (same total, different distribution). No-CoT worst. 7 hard Qs unsolved by any prompt.

Figure 8: Per-question correctness comparison across three prompting strategies (No-CoT, Zero-shot CoT, Few-shot CoT) on GSM8K Q1–300 (Qwen-Plus). A $\times$ indicates an incorrect answer. The 6 questions marked as shared failures are incorrectly answered under all three strategies. Few-shot CoT and Zero-shot CoT each have 13 errors but differ in which specific questions they miss, illustrating the zero-sum redistribution effect described in Section 4.3.1.

Q#	Category	Error Reason	RF1 NoPrompt	RF2 +Unit1	RF3 +Unit2	RF4 +Unit3	Total Fixed	Prompt Fixed	Stability
Q147	Interp. Ambiguity	3 times more pieces: 4x vs 3x	FIX	FIX	FIX	FIX	3	3/3	Stable
Q183	Arithmetic Error	half age timing: used 30/2 instead of 20/2	FIX	FIX	FIX	FIX	3	3/3	Stable
Q232	Interp. Ambiguity	2 times more shells: 3x vs 2x interpretation	FIX	FIX	FIX	FIX	3	3/3	Partial
Q294	Interp. Ambiguity	2 times more + 4th round bonus confusion	FIX	FIX	FIX	FIX	2	2/3	Partial
Q112	Missed Detail	boy-2-girl-1 free: strategy not fully utilized	FIX	FIX	FIX	FIX	2	2/3	Partial
Q1068	Unit/Concept Confusion	cents vs dollars output (245 vs 24500)	FIX	FIX	FIX	FIX	2	2/3	Partial
Q126	Interp. Ambiguity	used spoons: consumed vs still available	FIX	FIX	FIX	FIX	2	1/3	Unstable
Q41	Interp. Ambiguity	two times older than: 3x vs 2x	FIX	FIX	FIX	FIX	1	1/3	Unstable
Q242	Interp. Ambiguity	collectively read: unique books vs sum of reads	FIX	FIX	FIX	FIX	1	1/3	Unstable
Q188	Over-reasoning	simple vs compound interest confusion	FIX	FIX	FIX	FIX	1	1/3	Unstable
Q1188	Over-reasoning	snowball: per-minute net rate over-complication	FIX	FIX	FIX	FIX	1	1/3	Unstable
Q1039	Logic Chain Break	opportunity cost inclusion in: paycheck calculation	FIX	FIX	FIX	FIX	2	1/3	Unstable
Q467	Unit/Concept Confusion	dollar vs cents output confusion	FIX	FIX	FIX	FIX	1	1/3	Unstable

Figure 9: Per-question recovery status across four evaluation rounds for the 49 initially incorrect GSM8K problems. RF1 uses the original Zero-shot CoT prompt; RF2–RF4 append a unit-conversion reminder. A $\checkmark$ indicates a correct answer in that round; $\times$ indicates an incorrect answer. The “Fix Count” column shows the number of prompt-augmented runs (RF2–RF4) in which the question was correctly answered. Only 2 questions (Q147, Q183) achieve a fix count of 3/3, while 40 questions never recover (0/3), confirming the high stochasticity reported in Section 4.2.4.

Q#	Error Description	NoPrompt	ZeroShot	FewShot	Freq
#137	wrong calc: 21 vs 11 (3/3 runs)	No Prompt	Zero-shot	Few-shot	3/3
#197	wrong calc: 71 vs 81 (3/3 runs)	No Prompt	Zero-shot	Few-shot	3/3
#92	very wrong: 799/5345 vs 167	---	Zero-shot	Few-shot	2/3
126	off by 1-6: 27/20 vs 26	---	Zero-shot	Few-shot	2/3
#120	wrong var: 4p vs b=4	No Prompt	---	Few-shot	2/3
#21	wrong sum: 10 vs 18	---	---	Few-shot	1/3
#46	wrong real part: 12+9i vs 6+9i	---	---	Few-shot	1/3
#51	wrong value: 6 vs 4	---	---	Few-shot	1/3
#74	wrong calc: 69/4 vs 19/4	No Prompt	---	---	1/3
#86	wrong calc: 497 vs 129	No Prompt	---	---	1/3
#108	off by 0.5: 47.5 vs 47	No Prompt	---	---	1/3
#122	wrong calc: 13 vs 7	---	---	Few-shot	1/3
#201	wrong calc: -25 vs -120	No Prompt	---	---	1/3

Figure 10: Persistent reasoning errors of Qwen-Turbo on MATH-Algebra (Q1–300) across three prompting strategies. A $-$ indicates an correct answer under that strategy. Problems #137 and #197 fail universally across all three configurations, indicating hard reasoning deficits. Problems #21, #46, #51, and #122 are uniquely missed by Few-shot CoT, providing direct evidence for the cognitive overload hypothesis discussed in Section 4.3.2(b).